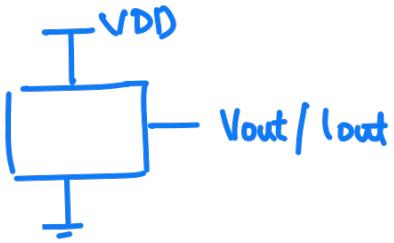
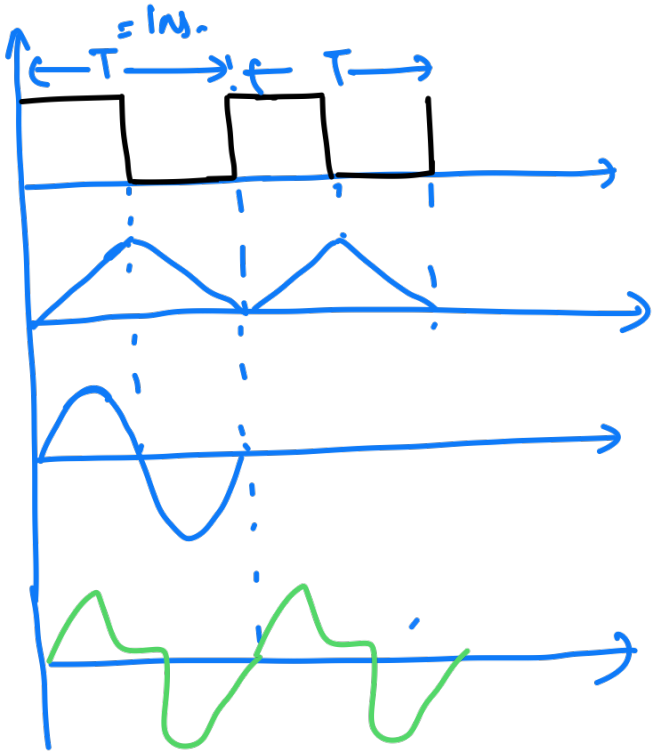
Lecture #33

Oscillators



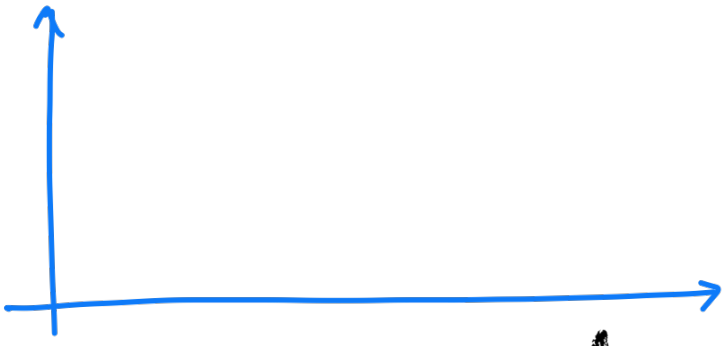


Clock: Digital circuits (Processor)

Sampling < ADCs
DACs

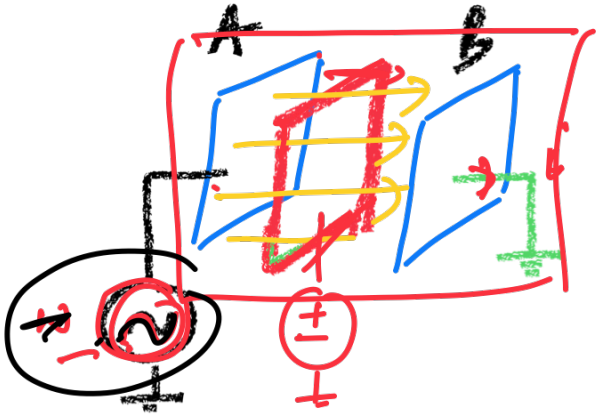
RF communication

Wired communication



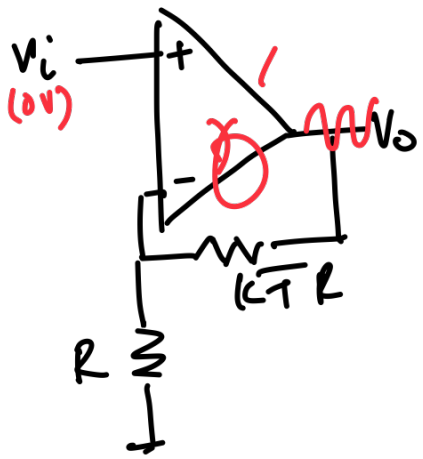
Reference clock.

- Atomic clock
- Quartz crystal



MEMS

- High frequency periodic systems.
  - Linear systems ✓
  - Non-linear system.



$$\frac{V_o}{V_i} = K.$$

$$A(s) = \frac{A_0}{(1+s/\omega_p)^3}$$

Loop gain, $LG = \dfrac{A_0 \beta K}{(1+s/\omega_p)^3}$

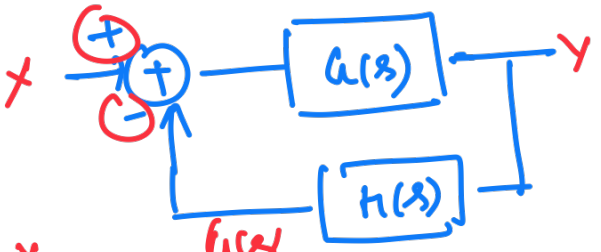
$$\frac{V_o}{V_i} = \frac{\quad}{1+LG}$$

Barkhausen criteria: ✓ $|LG(j\omega_{osc})| = 1$ ✓ } w/ neg. feedback.

$\angle LG(j\omega_{osc}) = (2K+1)\pi$

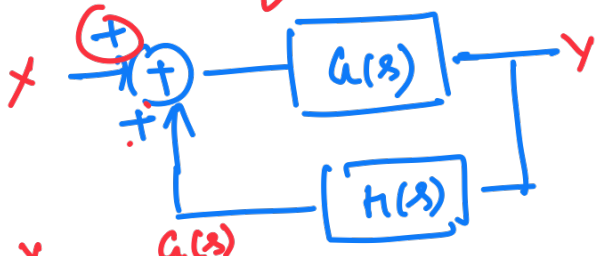
$|LG(j\omega_{osc})| = 1$ } w/ no negative sign in fb.

$\angle LG(j\omega_{osc}) = (2K)\pi$



$$\frac{Y}{X} = \frac{G(s)}{1+LG} \quad | \quad LG = G(s)H(s)$$

$$\left.\frac{Y}{X}\right|_{\omega=\omega_{osc}} = \frac{G(s)}{1+1\angle\pi} = \infty$$



$$\frac{Y}{X} = \frac{G(s)}{1-G(s)H(s)} \quad | \quad LG = G(s)H(s)$$

$$\left.\frac{Y}{X}\right|_{\omega=\omega_{osc}} = \frac{1}{1-1\angle 2\pi} = \infty$$

$$L_u(s) = \frac{A_0/K}{(1+s/\omega_p)^3}$$

$$\angle L_u = -3\tan^{-1}\left(\frac{\omega}{\omega_p}\right)$$

$|L_u(j\omega_{osc})| = 1$, $\angle L_u(j\omega_{osc}) = (2k+1)\pi = -\pi, 0, +\pi, 3\pi, \ldots$

$$\angle L_u = -3\tan^{-1}\left(\frac{\omega_{osc}}{\omega_p}\right) = -180^\circ$$

$$\Rightarrow \frac{\omega_{osc}}{\omega_p} = \tan(60^\circ) = \sqrt{3}$$

$$\boxed{\omega_{osc} = \sqrt{3}\,\omega_p}$$

$$\left[\frac{A_0/K}{\left(\sqrt{1+\left(\frac{\sqrt{3}\omega_p}{\omega_p}\right)^2}\right)^3}\right] = 1 \quad \Rightarrow \boxed{\frac{A_0}{K} = 8}$$

$$\frac{V_o}{V_i} = \frac{\quad}{1+L_u}$$

$$D(s) = 1+L_u = 1 + \frac{A_0/K}{(1+s/\omega_p)^3}$$

$$= \frac{\left(1+\frac{s}{\omega_p}\right)^3 + \frac{A_0}{K}}{(\quad)} = 0$$

$$\left(1+\frac{s}{\omega_p}\right)^3 + \frac{A_0}{K} = 0$$

$$\left(1+\frac{s}{\omega_p}\right)^3 = (-1)\frac{A_0}{K} = e^{j(2k+1)\pi}\frac{A_0}{K}$$

$$\frac{s}{\omega_p} = -1 + e^{j(2k+1)\pi/3}\cdot\left(\frac{A_0}{K}\right)^{1/3}$$

$$= -1 + 2\,e^{j(2k+1)\pi/3}$$

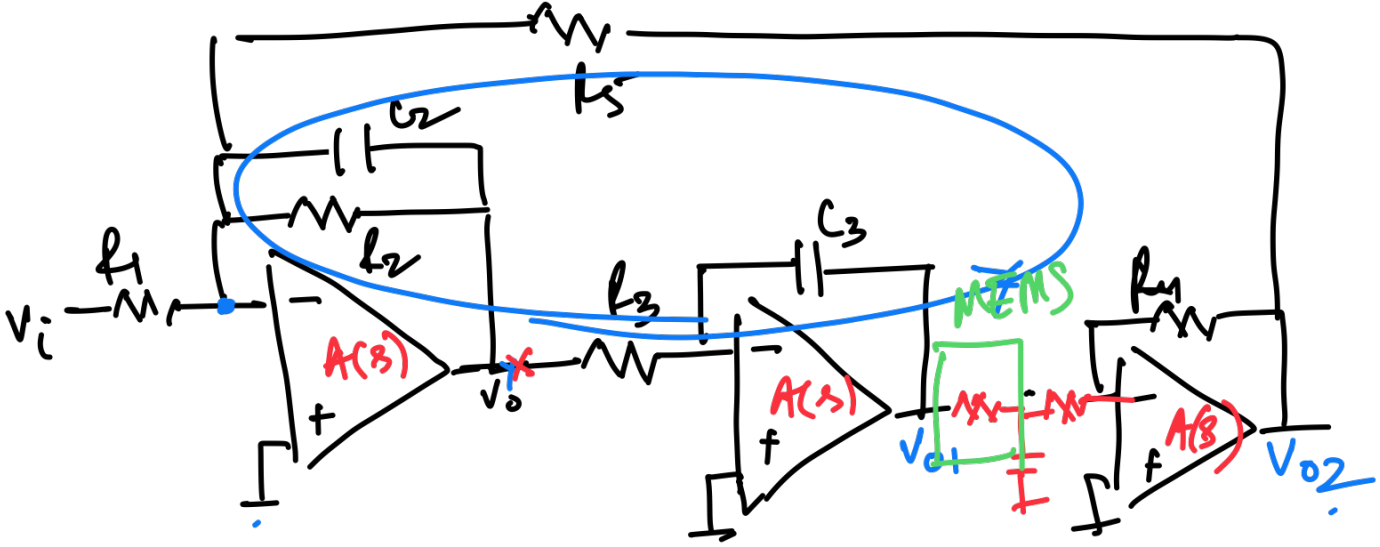
$$= \left(-1+2e^{j\pi/3}\right),\ \left(-1+2e^{j\pi}\right),\ \left(-1+2e^{j5\pi/3}\right)$$

$$= -1+2\left(\frac{1}{2}+j\frac{\sqrt{3}}{2}\right),\ -1+2(-1),\ -1+2\left(\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)$$

$$= +j\sqrt{3},\ -3,\ -j\sqrt{3}$$

$$s = j\sqrt{3}\,\omega_p,\ -3\omega_p,\ -j\sqrt{3}\,\omega_p$$





$$LG(s) = \frac{-1}{sC_3R_3} \times -1 \times \frac{1}{R_5} \times -\left(R_2 \| \frac{1}{sC_2}\right)$$

$$= \frac{-1}{sC_3R_3} \times \frac{1}{R_5} \frac{R_2}{1+sC_2R_2}$$

$$= \frac{-R_2/5}{sC_3R_3\,(1+sC_2R_2)}$$